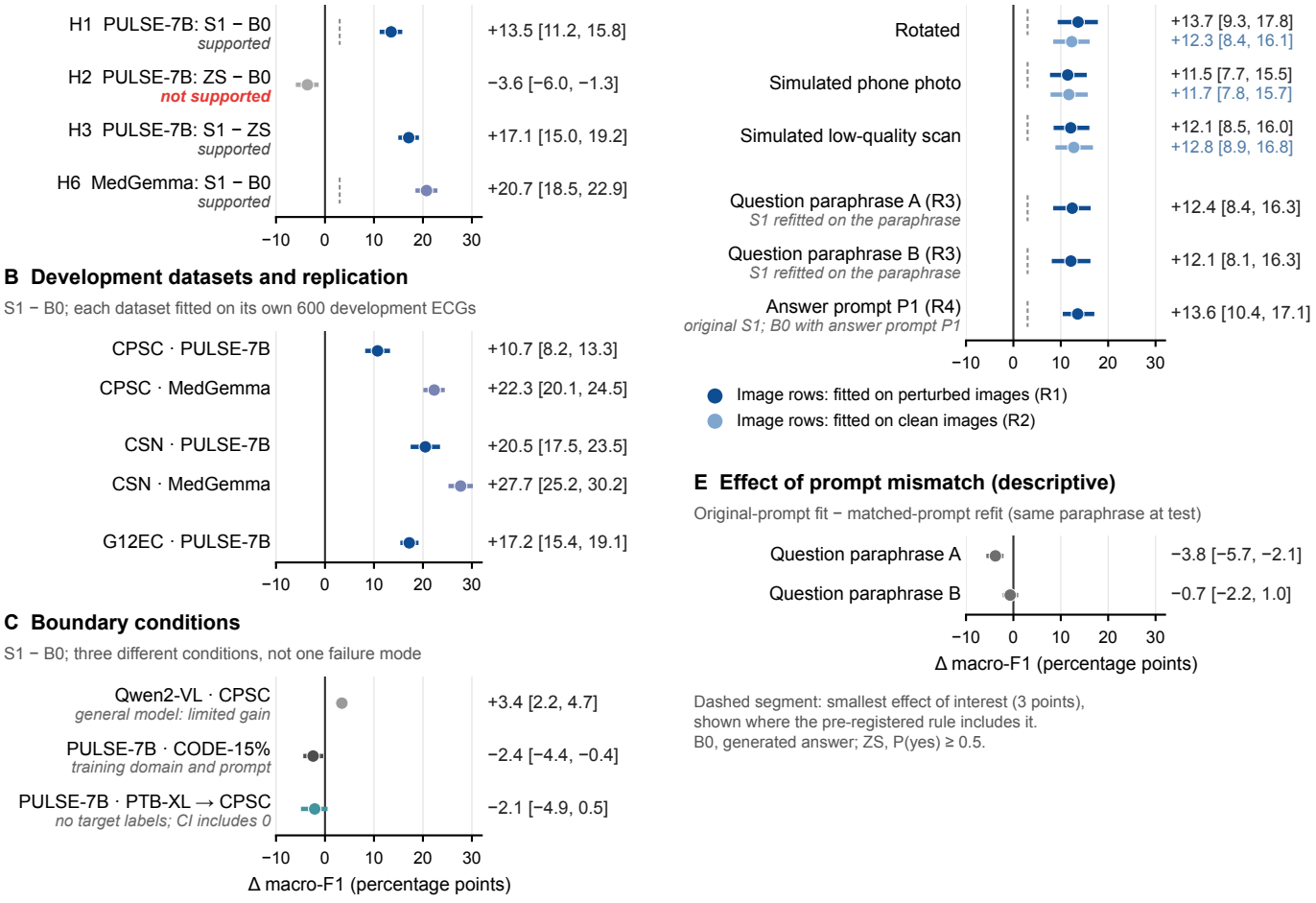

A SPH validation, pre-registered
2312 test ECGs; verdict under each row from its pre-registered rule
H1 PULSE-7B: S1 − B0
supported
+13.5 [11.2, 15.8]
H2 PULSE-7B: ZS − B0
not supported
−3.6 [−6.0, −1.3]
H3 PULSE-7B: S1 − ZS
supported
+17.1 [15.0, 19.2]
H6 MedGemma: S1 − B0
supported
+20.7 [18.5, 22.9]
−10 0 10 20 30
B Development datasets and replication
S1 − B0; each dataset fitted on its own 600 development ECGs
CPSC · PULSE-7B
+10.7 [8.2, 13.3]
CPSC · MedGemma
+22.3 [20.1, 24.5]
CSN · PULSE-7B
+20.5 [17.5, 23.5]
CSN · MedGemma
+27.7 [25.2, 30.2]
G12EC · PULSE-7B
+17.2 [15.4, 19.1]
−10 0 10 20 30
C Boundary conditions
S1 − B0; three different conditions, not one failure mode
Qwen2-VL · CPSC
general model: limited gain
+3.4 [2.2, 4.7]
PULSE-7B · CODE-15%
training domain and prompt
−2.4 [−4.4, −0.4]
PULSE-7B · PTB-XL → CPSC
no target labels; CI includes 0
−2.1 [−4.9, 0.5]
−10 0 10 20 30
Δ macro-F1 (percentage points)
D Robustness, pre-registered
S1 − B0; PULSE-7B; 300 fitting / 800 test SPH ECGs; all 9 rules met
Rotated
+13.7 [9.3, 17.8]
+12.3 [8.4, 16.1]
Simulated phone photo
+11.5 [7.7, 15.5]
+11.7 [7.8, 15.7]
Simulated low-quality scan
+12.1 [8.5, 16.0]
+12.8 [8.9, 16.8]
Question paraphrase A (R3)
S1 refitted on the paraphrase
+12.4 [8.4, 16.3]
Question paraphrase B (R3)
S1 refitted on the paraphrase
+12.1 [8.1, 16.3]
Answer prompt P1 (R4)
original S1; B0 with answer prompt P1
+13.6 [10.4, 17.1]
−10 0 10 20 30
Image rows: fitted on perturbed images (R1)
Image rows: fitted on clean images (R2)
E Effect of prompt mismatch (descriptive)
Original-prompt fit − matched-prompt refit (same paraphrase at test)
Question paraphrase A
−3.8 [−5.7, −2.1]
Question paraphrase B
−0.7 [−2.2, 1.0]
−10 0 10 20 30
Δ macro-F1 (percentage points)
Dashed segment: smallest effect of interest (3 points),
shown where the pre-registered rule includes it.
B0, generated answer; ZS, P(yes) ≥ 0.5.